STA6166 Statistical Methods I

Final Project: Multiple Linear Regression and Variable Selection

Total: 100 Points

Student Name: _____ Date: _____

Project Overview

You will analyze a dataset containing one continuous response variable Y and 15 candidate predictors X_1, …, X_{15}. The data were generated from a multiple linear regression model in which exactly 5 predictors have nonzero population regression coefficients; the other 10 have coefficients equal to zero. The identities of the 5 true predictors will not be provided.

Your tasks are to fit the full regression model, examine its assumptions, select predictors using one classical model-selection approach, fit a LASSO regression with a cross-validated tuning parameter, and compare the resulting models. Complete the analysis in R. Unless otherwise stated, use a 5% significance level.

Part 1. Full Multiple Linear Regression and Diagnostics (20 Points)

Question 1. Fit the full model (5 points). Fit the model $Y_i = \beta_0 + \sum_{j=1}^{15} \beta_j X_j + \epsilon_i$ Report the coefficient estimates, standard errors, t-statistics, and p-values. Include the model's R^2 and adjusted R^2.

Fill in the following table using the R output from the full model with 15 predictors.

Coefficient	Point estimate	Standard error	t-statistic	p-value
β_1	**1.9115**	**0.7175**	**2.664**	**0.009253**
β_2	0.2025	0.6982	0.290	0.772483
β_3	-0.3949	0.6771	-0.583	0.561335
β_4	**-2.6473**	**0.8055**	**-3.287**	**0.001481**
β_5	1.1445	0.7498	1.526	0.130681
β_6	0.9065	0.7347	1.234	0.220698
β_7	1.2767	0.6619	1.929	0.057121
β_8	0.2487	0.7648	0.325	0.745818
β_9	-0.1041	0.6638	-0.157	0.875731
β_{10}	**7.2157**	**0.7176**	**10.056**	**4.49e-16**

β_{11}	-0.1946	0.6258	-0.311	0.756631
β_{12}	0.3334	0.6508	0.512	0.609722
β_{13}	**-2.6675**	**0.6812**	**-3.916**	**0.000183**
β_{14}	**-1.6059**	**0.6871**	**-2.337**	**0.021801**
β_{15}	0.5023	0.6974	0.720	0.473400

Rows in bold are significant at α = 0.05.

> The full model explains about 65% of the variation in Y: **R^2 = 0.6506** and **adjusted R^2 = 0.5882**. It is highly significant overall (F = 10.43 on 15 and 84 df, p = 1.442e-13). Five slopes are significant at the 5% level (**X1, X4, X10, X13, X14**). X7 is borderline (p = 0.057), and the other nine predictors are not significant. The intercept is 1.5842 (SE = 0.7122, t = 2.224, p = 0.0288).

```
> summary(full_model)     # last lines of the output
Residual standard error: 6.427 on 84 degrees of freedom
Multiple R-squared:  0.6506,    Adjusted R-squared:  0.5882
F-statistic: 10.43 on 15 and 84 DF,  p-value: 1.442e-13
```

Question 2. Unequal error variances (5 points). Apply the Breusch–Pagan test to the full regression model.

(a) State the null and alternative hypothesis.

> **H_0:** the error variance is constant, $Var(\varepsilon_i \mid X_i) = \sigma^2$ for all i (homoscedasticity).
>
> **H_1:** the error variance depends on the predictors $X_1, \ldots, X_{15}$ (heteroscedasticity).

(b) Write down the test statistic and p-value you get from the R function.

> **BP = 17.673** on 15 df, **p-value = 0.2802**.

```
> bptest(full_model)
    studentized Breusch-Pagan test

data:  full_model
BP = 17.673, df = 15, p-value = 0.2802
```

(c) State if you have enough evidence to reject H_0.

> **No.** Since p = 0.2802 > 0.05, we fail to reject H_0. There is no significant evidence of unequal error variances, so the constant-variance assumption is reasonable.

Question 3. Normality diagnostic (5 points). Apply the Shapiro–Wilk test to test the normality of residuals of the full regression model.

(a) State the null and alternative hypothesis.

> **H_0:** the regression errors are normally distributed.
>
> **H_1:** the regression errors are not normally distributed.

(b) Write down the test statistic and p-value you get from the R function.

> **W = 0.99144**, **p-value = 0.7795**.

```
> shapiro.test(residuals(full_model))
    Shapiro-Wilk normality test

data:  res
W = 0.99144, p-value = 0.7795
```

(c) State if you have enough evidence to reject H_0.

> **No.** Since p = 0.7795 > 0.05, we fail to reject H_0. The residuals show no evidence of non-normality. (OLS residuals are slightly correlated and have unequal variances, so this p-value is a descriptive check rather than an exact test.)

Question 4. Diagnostic plots (5 points). Provide (a) a residuals-versus-fitted plot and (b) a normal Q–Q plot. Briefly discuss whether the plots are consistent with constant error variance and normality.

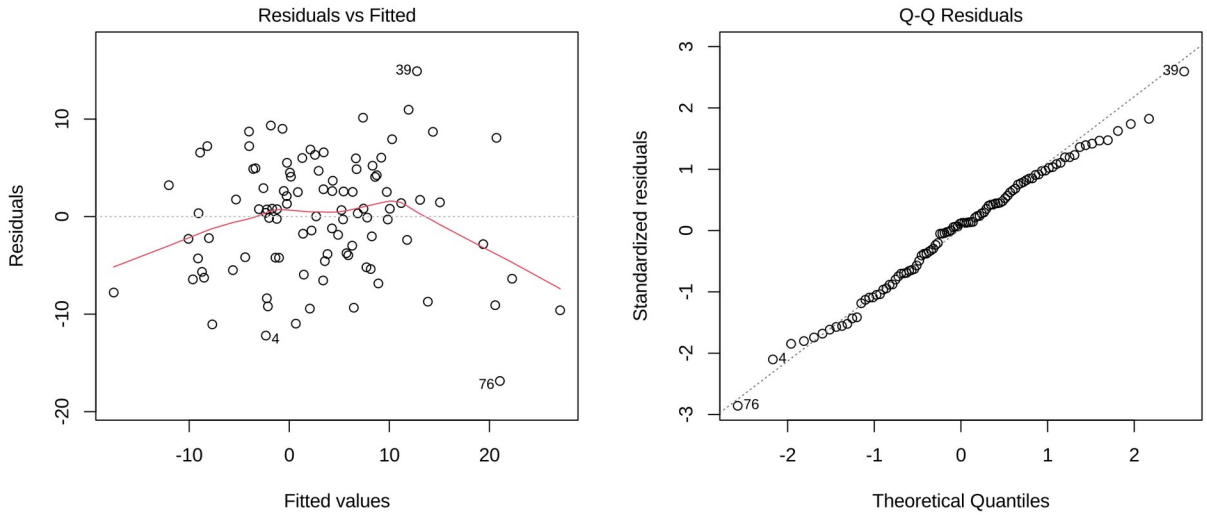


Figure 1. (a) Residuals versus fitted values and (b) normal Q–Q plot for the full model.

> **(a) Residuals vs fitted.** The points scatter randomly around zero, with roughly constant spread across the fitted values and no funnel shape. The red smoother dips at both ends, but only because of a few points at extreme fitted values (e.g., observation 76), not a systematic curve. This supports constant variance and agrees with the Breusch–Pagan test.
>
> **(b) Normal Q–Q.** The standardized residuals follow the reference line closely, with only small departures in the tails (observations 39, 76 and 4). This supports normal errors and agrees with the Shapiro–Wilk test.
>
> **Conclusion:** the constant-variance and normality assumptions of the full model are reasonable.

Part 2. Classical Variable Selection (30 Points)

Question 5. Choose a search procedure (5 points). Choose ONE procedure: forward selection, backward elimination, or bidirectional stepwise selection. Explain how the procedure (the ONE procedure you choose) adds or removes variables.

> **Chosen procedure: bidirectional stepwise selection.**

- **Start:** the intercept-only model, Y ~ 1.
- **Each step:** try every single **addition** of a predictor not in the model and every single **removal** of a predictor in the model, then make the move that lowers the criterion most.
- **Why both directions:** a predictor added early can be dropped later if others make it redundant, which pure forward selection cannot do.
- **Stop:** when no single addition or removal improves the criterion.

```
> step(null_model,
+      scope = list(lower = formula(null_model), upper = formula(full_model)),
+      direction = "both", k = log(N), trace = 0)
```

Question 6. Choose one selection criterion (5 points). Choose exactly one criterion: AIC, BIC, Mallows' C_p, or adjusted R^2. Explain what the criterion measures and why you chose it. Once selected, keep this criterion fixed throughout Part 2; do not switch criteria after seeing the results.

For AIC or BIC, you may use stats::step() with forward, backward, or bidirectional search. For C_p or adjusted R^2, you may use leaps::regsubsets() with forward or backward search. The seqrep method in regsubsets() is sequential replacement, not bidirectional stepwise regression. If you choose an unsupported combination, you must implement and document the search algorithm yourself. You may revise your initial choice of search procedure before running the analysis to use one of the supported combinations.

> **Chosen criterion: BIC (Bayesian Information Criterion), minimized.**
>
> BIC = −2 log L + p·log(N), where L is the maximized likelihood and p the number of parameters; for a Gaussian linear model this equals N·log(RSS/N) + p·log(N) plus a constant. The first term measures **lack of fit** and the second penalizes **complexity**. A lower BIC means a better balance.
>
> **Why BIC:**

- Its penalty is log(100) ≈ 4.61 per parameter, more than twice AIC's 2, so it favours parsimonious models.
- It is **consistent**: if the true model is among the candidates, BIC picks it with probability approaching 1 as N grows. That suits this project, whose aim is to recover a sparse set of truly nonzero predictors.
- Bidirectional search with BIC is supported directly by stats::step(), so no custom algorithm is needed.

> The criterion was fixed before the analysis was run.

Question 7. Select predictors (10 points). Apply the chosen search procedure and fixed criterion.

(a) List the predictors in the final selected model.

> **X1, X4, X10, X13 and X14.**

(b) Report the final model formula.

> **Y ~ X10 + X4 + X13 + X14 + X1** (in order of entry), i.e.

$\hat{Y} = 1.9307 + 1.6604 \cdot X1 - 2.6507 \cdot X4 + 7.2586 \cdot X10 - 2.3389 \cdot X13 - 1.8656 \cdot X14$

```
> formula(stepwise_BIC)
Y ~ X10 + X4 + X13 + X14 + X1
```

(c) Report the chosen criterion's value for the selected model.

BIC = 680.8865, down from 752.8193 for the intercept-only model.

```
> BIC(stepwise_BIC)
[1] 680.8865
```

(d) State the number of selected predictors, excluding the intercept. Briefly describe how you identified the final model along the search path.

5 predictors (intercept excluded).

Starting from Y ~ 1, the search added **X10 → X4 → X13 → X14 → X1**, and the criterion fell at every step (BIC 752.82 → 693.19 → 689.44 → 686.18 → 682.27 → 680.89). At each step it also checked every removal, but none ever helped. At the final model, the best remaining addition (X7) and every removal would raise the criterion, so the search stopped there.

```
> stepwise_BIC$anova    # search path ("AIC" column is BIC here: k = log(N))
   Step Df  Deviance Resid. Df Resid. Dev      AIC
1       NA        NA        99   9930.664 464.4264
2 + X10 -1 4706.6253        98   5224.039 404.7959
3  + X4 -1  418.8081        97   4805.231 401.0445
4 + X13 -1  363.4433        96   4441.787 397.7849
5 + X14 -1  362.3603        95   4079.427 393.8800
6  + X1 -1  237.2439        94   3842.183 392.4936
```

Robustness check. The R script also ran every search direction with every criterion (code section 6). Forward, backward and bidirectional searches agree for each criterion, and BIC gives the smallest model:

Table A. Classical selection results for all 12 search/criterion combinations.

Criterion	Selected predictors (forward = backward = stepwise)	No.	Value
AIC	X1, X4, X5, X7, X10, X13, X14	7	659.6584
BIC (chosen)	**X1, X4, X10, X13, X14**	**5**	**680.8865**
Mallows' Cp	X1, X4, X5, X7, X10, X13, X14	7	2.7414
Adjusted R^2	X1, X4, X5, X6, X7, X10, X13, X14	8	0.6130

Question 8. Examine the selected model (10 points). Refit an ordinary least-squares model using the selected predictors.

(a) Fill in the following table using the R output from the model with selected predictors. If a variable is not selected, leave it empty.

Coefficient	Point estimate	Standard error	t-statistic	p-value
β_1	1.6604	0.6892	2.409	0.017936
β_2				
β_3				
β_4	-2.6507	0.7873	-3.367	0.001103

β_5				
β_6				
β_7				
β_8				
β_9				
β_{10}	7.2586	0.6732	10.782	< 2e-16
β_{11}				
β_{12}				
β_{13}	-2.3389	0.6343	-3.687	0.000379
β_{14}	-1.8656	0.6545	-2.850	0.005366
β_{15}				

(b) Report adjusted R^2.

> **Adjusted R^2 = 0.5925** (R^2 = 0.6131).

```
> summary(stepwise_BIC)     # last lines of the output
Residual standard error: 6.393 on 94 degrees of freedom
Multiple R-squared:  0.6131,    Adjusted R-squared:  0.5925
F-statistic: 29.79 on 5 and 94 DF,  p-value: < 2.2e-16
```

(c) Compare its adjusted R^2 with that of the full model.

> The selected model's adjusted R^2 (0.5925) is **slightly higher** than the full model's (0.5882), even though it uses 10 fewer predictors. R^2 itself falls from 0.6506 to 0.6131, as it always does when terms are removed. Once the cost of extra parameters is accounted for, however, the ten dropped predictors were not worth keeping.

Table B. Full versus selected model.

Model	Predictors	R^2	Adjusted R^2	Residual SE (df)
Full model	15	0.6506	0.5882	6.427 (84)
Selected (BIC)	**5**	**0.6131**	**0.5925**	**6.393 (94)**

(d) Discuss whether the selected model is simpler and what information may have been lost by excluding predictors.

> **Simpler:** yes, 5 predictors instead of 15, and every retained coefficient is significant at 5% (largest p = 0.0179). The estimates barely change from the full model, and all five standard errors become slightly smaller.
>
> **What may be lost:** the ten excluded predictors explained about 3.75 percentage points of R^2 (RSS rises from 3469.5 to 3842.2). A partial F-test shows this loss is not significant (F = 0.9023, p = 0.5349). Still, if an excluded predictor has a small real effect, that effect is now pushed into the error term and can no longer

be estimated. X7 ($p = 0.057$ in the full model; kept by AIC and Cp) and X5 are the likeliest candidates. Because the predictors are only weakly correlated (largest $|r| = 0.295$; all VIFs < 1.25), dropping them causes little bias in the remaining estimates.

```
> anova(stepwise_BIC, full_model)     # additional check: partial F-test
  Res.Df    RSS Df Sum of Sq      F Pr(>F)
1     94 3842.2
2     84 3469.5 10    372.69 0.9023 0.5349
```

Part 3. LASSO Regression (30 Points)

Question 9. Explain LASSO (5 points). Consider the LASSO objective: min over β_0, β of

$$\frac{1}{2N}\sum_{i=1}^{N}\left(Y_i-\beta_0-\sum_{j=1}^{15}X_{ij}\beta_j\right)^2+\lambda\sum_{j=1}^{15}\left|\beta_j\right|.$$

(a) Explain the role of λ.

> λ ≥ 0 is the **tuning parameter** that sets the weight of the L1 penalty λ·Σ|βⱼ|, i.e. the trade-off between fitting the data and keeping coefficients small:
>
> - λ = 0 gives ordinary least squares.
> - As λ increases, all coefficients shrink toward zero and more become exactly zero: lower variance, more bias.
> - For λ ≥ λ_max (≈ 6.86 here), every slope is zero and only the intercept remains.

(b) Explain why LASSO can produce exactly zero coefficient estimates. The intercept is not penalized.

> The absolute-value penalty has a **corner at zero** (it is not differentiable there). With standardized predictors each coefficient is updated by **soft-thresholding**:
>
> estimate of β_j = sign(z_j)·max(|z_j| − λ, 0), where z_j = (1/N)·Σᵢ x_{ij}·(partial residual)ᵢ.
>
> Whenever a predictor's association with the residual satisfies |z_j| ≤ λ, its estimate is exactly 0, not just small. Geometrically, the constraint region Σ|βⱼ| ≤ t is a diamond with corners on the axes, and the RSS contours usually touch it first at a corner, where some coefficients are zero. Ridge regression's round region has no corners, so it shrinks but never zeroes. The intercept β_0 is not penalized, so it is never shrunk.

Question 10. Choose λ by cross-validation (10 points). Use glmnet::cv.glmnet() with 10-fold cross-validation and Gaussian squared-error loss. Use a fixed random seed for reproducibility and standardize predictors using the package's default behavior.

(a) Include the cross-validation plot, plot the cross-validation MSE over λ.

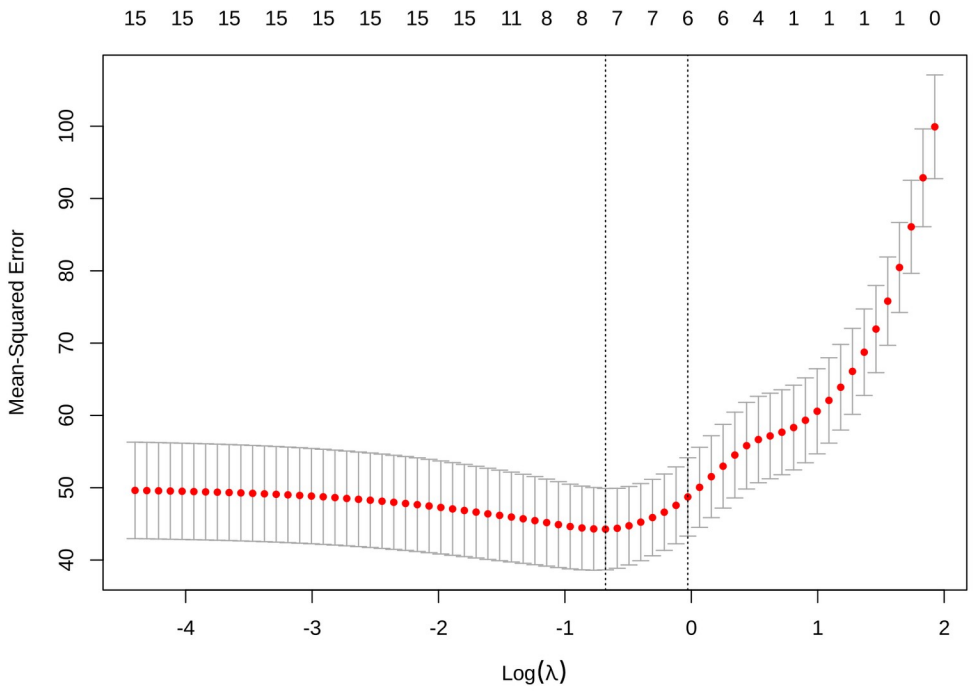


Figure 2. 10-fold cross-validated MSE versus log(λ) (seed 6166). Bars show ±1 SE; dashed lines mark lambda.min (left) and lambda.1se (right); the top axis gives the number of nonzero coefficients.

(b) Report lambda.min, the λ giving the smallest cross-validation MSE.

lambda.min = 0.5070394 (log λ ≈ −0.68), with minimum CV MSE = 44.2725 (SE = 5.628) and 8 nonzero coefficients.

```
> print(cv_lasso)

Call:  cv.glmnet(x = X, y = Y, type.measure = "mse", nfolds = 10, alpha = 1,
       family = "gaussian", standardize = TRUE)

Measure: Mean-Squared Error

    Lambda Index Measure    SE Nonzero
min 0.5070    29   44.27 5.628       8
1se 0.9725    22   48.73 5.419       6

> cv_lasso$lambda.min
[1] 0.5070394
```

(c) Explain how the cross-validation criterion selects λ. Use lambda.min throughout the remaining LASSO analysis; do not switch to lambda.1se.

- cv.glmnet() fits the LASSO path on a grid of λ values, decreasing from λ_max ≈ 6.86.
- With set.seed(6166), the 100 observations are randomly split into **10 folds**.
- For each fold, the model is fitted on the other 9 folds at every λ, and the squared prediction error is computed on the held-out fold.

- Averaging over the 10 folds gives **CV(λ)**, the estimated test MSE (red dots), with a standard error from the fold-to-fold variation (grey bars).
- **lambda.min** is the λ that minimizes CV(λ). Smaller λ overfits (high variance), and larger λ over-shrinks (high bias).

lambda.1se = 0.9725 (6 nonzero coefficients) is also marked on the plot but, as instructed, is not used; lambda.min is used throughout.

Question 11. Fit LASSO and identify variables (10 points). Using the selected λ, obtain the LASSO coefficient estimates.

(a) Report the coefficients.

Enter the LASSO point estimate for each predictor. If LASSO does not select a predictor, enter 0.

Coefficient	**LASSO point estimate**
β_1	**1.07944426**
β_2	0.00000000
β_3	0.00000000
β_4	**-1.92849719**
β_5	**0.41838736**
β_6	**0.06818692**
β_7	**0.89926976**
β_8	0.00000000
β_9	0.00000000
β_{10}	**6.62840784**
β_{11}	0.00000000
β_{12}	0.00000000
β_{13}	**-1.62217204**
β_{14}	**-1.16931949**
β_{15}	0.00000000

(b) List all predictors with nonzero coefficients.

X1, X4, X5, X6, X7, X10, X13, X14 (intercept = 1.6903, not counted).

```
> print(selected_predictors)
[1] "X1"  "X4"  "X5"  "X6"  "X7"  "X10" "X13" "X14"
```

(c) State how many predictors LASSO selected. Exclude the intercept from the count.

> LASSO selected **8 predictors**. The other 7 (X2, X3, X8, X9, X11, X12, X15) are set exactly to 0.

```
> length(selected_predictors)
[1] 8
```

Question 12. Interpret LASSO (5 points). Explain how the penalty affects the fitted coefficients and why some predictors are excluded. Discuss the relationship between the size of λ and model sparsity.

> **Effect on coefficients.** The penalty pulls every estimate toward zero compared with OLS, while keeping its sign. Weak predictors shrink proportionally much more than strong ones: X10 loses about 8%, X6 over 90%.

Table C. OLS versus LASSO estimates for the 8 predictors LASSO keeps.

Predictor	OLS (full model)	LASSO (lambda.min)	Shrinkage
X1	1.9115	1.0794	44%
X4	-2.6473	-1.9285	27%
X5	1.1445	0.4184	63%
X6	0.9065	0.0682	92%
X7	1.2767	0.8993	30%
X10	7.2157	6.6284	8%
X13	-2.6675	-1.6222	39%
X14	-1.6059	-1.1693	27%

> **Why some predictors are excluded.** A predictor is dropped when its contribution to reducing the squared error does not outweigh the penalty $\lambda|\beta_j|$; soft-thresholding then sets its estimate to exactly 0. The seven excluded predictors all had small, clearly non-significant OLS estimates ($|t| < 0.75$), and X6 only barely survives (0.068).
>
> **λ and sparsity.** Larger λ gives a sparser model. The top axis of Figure 2 shows the number of nonzero coefficients falling from 15 at small λ to 8 at lambda.min, 6 at lambda.1se, 1 (X10 only) for log λ between about 1 and 1.8, and 0 at $\lambda_{max} \approx 6.86$.

Part 4. Comparison and Discussion (20 Points)

Question 13. Compare selected predictors (10 points). Create a table with one row for each of X_1, …, X_{15} and two columns indicating whether each predictor was selected by (i) classical variable selection and (ii) LASSO. Report (a) the number selected by each method, (b) the predictors selected by both, and (c) those selected by only one method.

Predictor	Classical selection (Yes/No)	LASSO (Yes/No)
X1	**Yes**	**Yes**
X2	No	No
X3	No	No
X4	**Yes**	**Yes**
X5	No	**Yes**
X6	No	**Yes**
X7	No	**Yes**
X8	No	No
X9	No	No
X10	**Yes**	**Yes**
X11	No	No
X12	No	No
X13	**Yes**	**Yes**
X14	**Yes**	**Yes**
X15	No	No

> **(a)** Classical selection (stepwise BIC): **5** predictors. LASSO (lambda.min): **8** predictors.
>
> **(b)** Selected by both: **X1, X4, X10, X13, X14**.
>
> **(c)** Selected by only one method: **X5, X6, X7**, all by LASSO only. No predictor is chosen by the classical method alone, so the classical set is a subset of the LASSO set.

Question 14. Explain differences (5 points). Do the methods select the same predictors? Which selects fewer? Explain at least two reasons the selected sets may differ, considering the criteria, coefficient shrinkage, and correlation among predictors.

> **No.** Both methods keep the same five core predictors, but LASSO adds X5, X6 and X7, so the **classical BIC model is smaller** (5 vs 8). Reasons:
>
> - **Different criteria.** BIC charges log(100) ≈ 4.61 per parameter and aims at the true sparse model. lambda.min instead minimizes prediction error, which tends to keep weak predictors that lower MSE slightly. The same pattern appears within the classical methods: AIC and Cp keep 7 predictors, and adjusted R^2 keeps 8, exactly the LASSO set (Table A).
> - **Coefficient shrinkage.** LASSO lets a weak predictor in cheaply, with a tiny coefficient (X6 = 0.068, X5 = 0.418). Because lambda.min also shrinks the strong effects (X13: −2.67 → −1.62), the extra small

terms partly make up for that. In subset selection a predictor is either out or in at its full OLS size, paying the full BIC penalty.

- **Correlation among predictors.** Correlations here are weak (largest |r| = 0.295, X5 with X13; VIFs 1.05–1.24), so this is a minor factor. Still, X5 overlaps with X13 and X14 (r = 0.295, −0.211): once those are in, X5 adds little and greedy stepwise never adds it, whereas LASSO spreads the effect and keeps X5 with a small coefficient.
- **Search and randomness.** Stepwise selection is a greedy, discrete search, while LASSO follows a continuous path. lambda.min also depends on the random fold split.

Question 15. Final discussion (5 points). State which method you prefer for this dataset and justify your choice using model simplicity, the selected predictors, and the results of your analysis. Do not assume that a selected predictor is necessarily a true predictor: the identities of the 5 nonzero population coefficients are unknown to you.

Preferred: the classical bidirectional stepwise model with BIC (X1, X4, X10, X13, X14).

Table D. Summary of the three models.

Model	Predictors	Adjusted R^2	10-fold CV MSE*
Full OLS	15	0.5882	49.92
Stepwise BIC (OLS)	**5**	**0.5925**	**42.32**
LASSO (lambda.min)	8	—	44.27

**Same 10 folds as cv.glmnet (seed 6166). The BIC figure is somewhat optimistic because its predictors were chosen on the full data.*

- **Simplicity:** 5 predictors instead of 8, all significant ($p \le 0.018$), so the model is easy to interpret.
- **Fit:** higher adjusted R^2 than the full model, and the dropped predictors are jointly non-significant (partial F p = 0.53).
- **Selected predictors:** its five predictors are exactly the ones both methods agree on. The three LASSO-only predictors have small shrunken coefficients and were not significant in the full model (p = 0.131, 0.221, 0.057).
- **Results:** it predicts at least as well as LASSO (CV MSE 42.3 vs 44.3), and forward, backward and bidirectional BIC searches all reach it. LASSO's set is less stable: lambda.1se would keep only 6 predictors.

Caveat: being selected does not prove a predictor is truly nonzero. X7 is borderline (the next variable stepwise BIC would add, and kept by AIC and Cp), so it may be a real but weak effect. If pure prediction were the goal, LASSO would be a sound alternative. For a small, interpretable set of predictors, the BIC model is the better choice.

Submission Requirements

Submit only the completed Microsoft Word document (.docx) containing your answers to all project questions. No additional files are required.

Grading Summary

Component	Points
Part 1. Full regression and diagnostics	20
Part 2. Classical variable selection	30
Part 3. LASSO regression	30
Part 4. Comparison and discussion	20
Total	100